Asteron Endowment: Assignment 4 Research Report

This report rebuilds the Assignment 3 SAA baseline on the full Zion history and evaluates the lowest-drawdown tactical overlay I could find under the Assignment 4 policy limits.

Strategic And Tactical Allocations

The tactical risk-off allocation below is the most effective overlay found in testing. When the TAA signal fires, the portfolio is re-optimized inside a tactical band of SAA weight +/-15% per asset class.

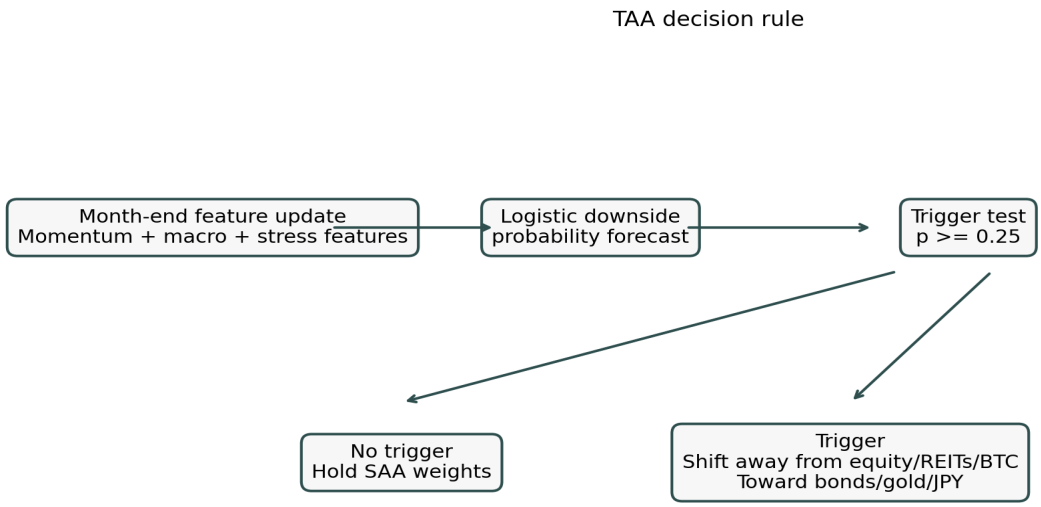
strategy	US Equity	US Bonds	REITs	Gold	Bitcoin	JPY
SAA baseline	30.0%	30.0%	15.0%	3.0%	0.0%	22.0%
TAA Strategy	24.6%	45.0%	0.8%	3.0%	0.1%	26.5%

1. Signal Design & Rationale

The final strategy uses a conservative logistic trigger of p >= 0.25. I tested logistic regression, SARIMAX, and XGBoost, then selected the specification that performed best once the wider tactical deviation band was allowed. The model uses recent market momentum, recent volatility, correlation across assets, gold and JPY strength, and macro stress measures such as VIX, the yield curve, and credit spreads. To train the model, I look at old data and mark each month as risky or not risky based on what happened to the SAA baseline in the following month. That next-month result is only used to train the model on past periods. When the strategy is actually making a decision, it only uses information available at that point in time. If the model gives a risk-off warning at year-end, the portfolio moves to the defensive TAA mix for the next period. In the final sample, risky months make up 15.0% of observations.
The practical signal objective is not to maximize forecast accuracy. It is to detect the months when the portfolio should shift into its most defensive feasible tactical allocation.

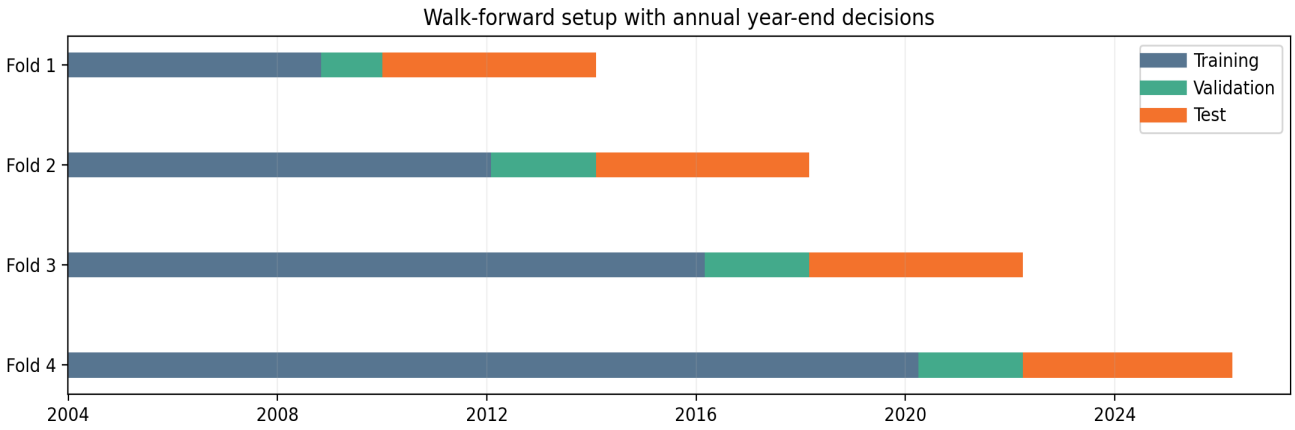
2. TAA Rule

The overlay only acts at each year-end rebalance. When the selected model flags elevated downside risk, the portfolio is re-solved as a minimum-variance allocation subject to SAA weight +/-15% per asset class.



3. Walk-Forward Setup

Four expanding folds are used on annual decision dates. Validation identifies one globally defensible rule, and that same annual overlay is then carried across the test folds.



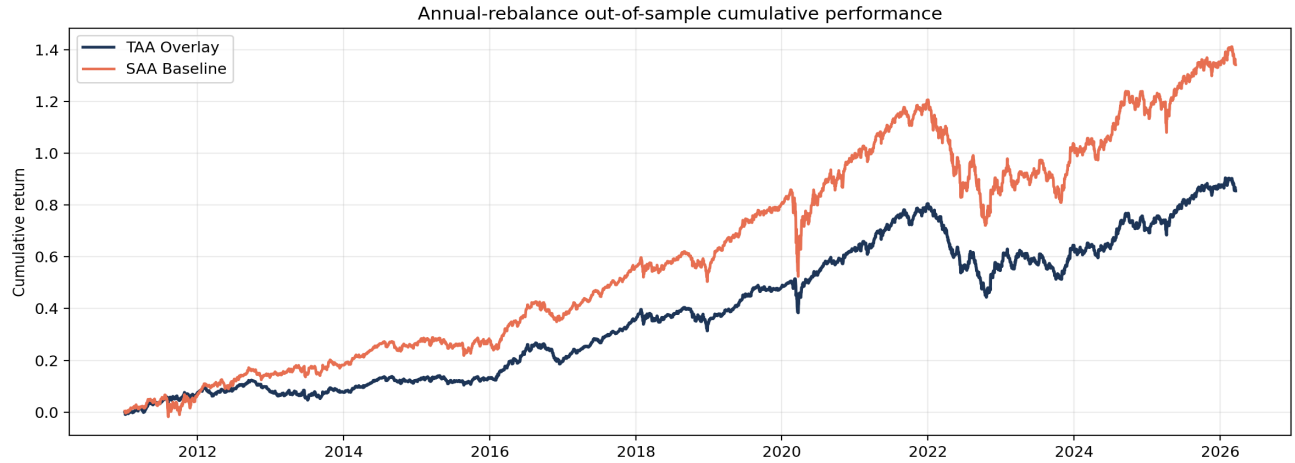
4. Training Results

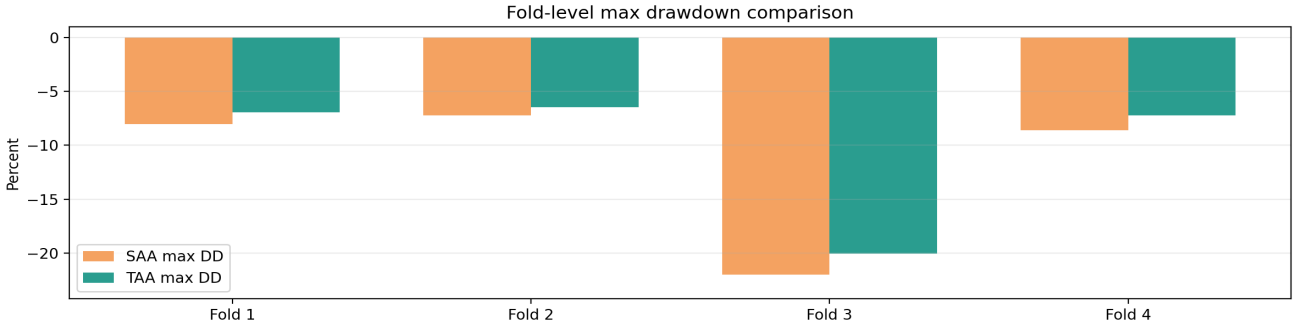
The table below shows the best-performing rule from each model family. Selection prioritized lower validation max drawdown rather than raw prediction accuracy.
The final strategy uses a conservative logistic trigger of p >= 0.25. This is the rule that produced the best downside result under the updated tactical deviation limits.

model_family	model_label	trigger_rule	validation_max_drawdown	validation_cagr	validation_trigger_rate
Logistic	Logistic regression	p >= 0.25	-9.5%	3.8%	87.5%
Ensemble	Logistic OR BayesXGBoost	logit >= 0.40 or xgb >= 0.25	-10.2%	3.9%	37.5%
BayesXGBoost	Bayes-tuned XGBoost	p >= 0.40	-12.3%	5.6%	0.0%
SARIMAX	SARIMAX(1,0,0)+macro	forecast < -0.2%	-12.3%	5.6%	0.0%

5. Test Results

All performance below is strictly out-of-sample across the walk-forward test folds.





fold	model	taa_max_drawdown	saa_max_drawdown	taa_return_pa	saa_return_pa
1	Logistic	-6.9%	-8.0%	2.7%	5.5%
2	Logistic	-6.5%	-7.2%	3.8%	4.2%
3	Logistic	-20.1%	-22.0%	2.7%	3.4%
4	Logistic	-7.2%	-8.6%	3.7%	5.0%

6. Overall Evaluation

The strategic benchmark below is the refreshed SAA baseline rather than the old 60/40 benchmark. The full strategic rebuild begins in 2003, while this comparison is restricted to the later out-of-sample tactical window.

portfolio	total_return_pa	volatility_pa	max_drawdown	historical_var_95_daily	calmar
TAA Overlay	3.2%	4.8%	-20.1%	-0.5%	0.161
SAA Baseline	4.5%	7.0%	-22.0%	-0.6%	0.203

IPS Scorecard

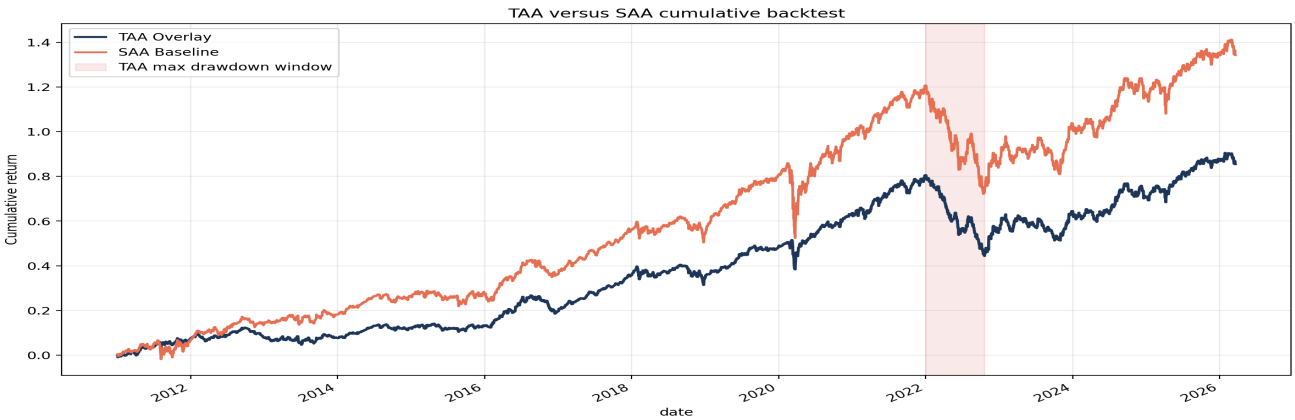
The table below compares the final TAA strategy against the Assignment 4 policy targets for return, volatility, and max drawdown.

metric	actual	target	result
Return p.a.	3.2%	5.0%	Fail
Volatility p.a.	4.8%	11.0%	Pass
Max drawdown	-20.1%	-13.0%	Fail

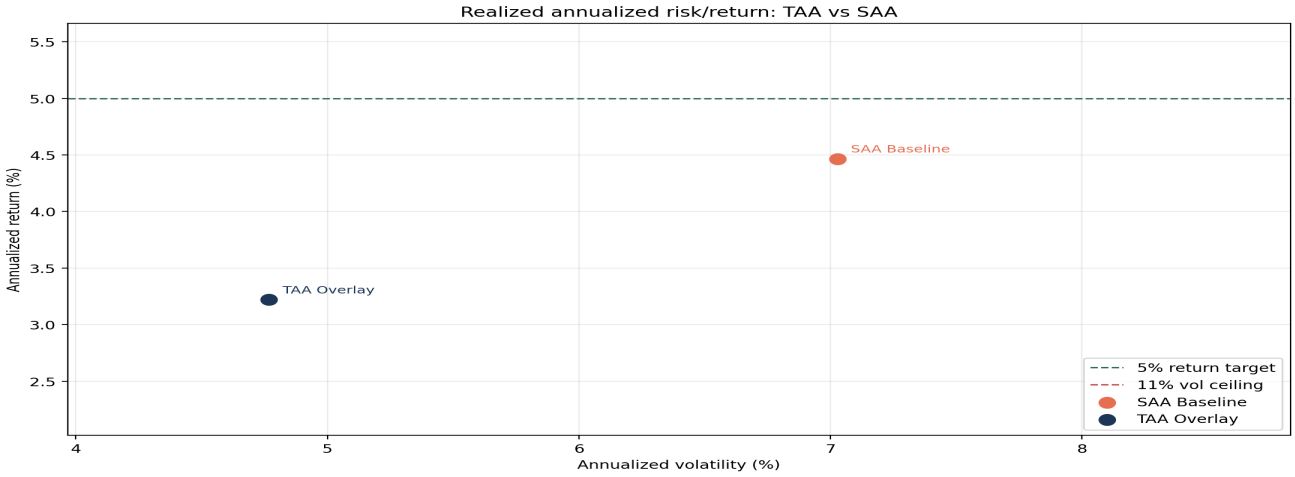
I had assumed that the 2008 financial crisis was the reason I was unable to achieve a max drawdown lower than 20%, so I ran the backtest again starting in 2010, which allowed me to avoid the financial crisis and the staggered portfolio system from the previous seven years. Even after doing that, max drawdown only improved from -26.3% to -22.0%, which shows that the recent rate-shock period is still the binding downside event.
Validation results should be interpreted as short-horizon screening evidence rather than as a full-cycle portfolio outcome. The realized out-of-sample TAA path still records a max drawdown of -20.1%, largely because the most severe stress episode in the tactical sample is the 2021-2022 inflation and rate-hike shock. In other words, the conservative logistic rule improves downside risk relative to the SAA baseline, but it does not eliminate the portfolio's exposure to the recent rate-shock regime.

Strategy Context

The chart below compares the cumulative out-of-sample paths of the final TAA strategy and the SAA baseline over the same window.



The next chart compares the realized annualized return and volatility of the final TAA strategy against the SAA baseline over the same out-of-sample window.



7. Recommendation

Recommendation: the tactical overlay improves downside risk, but not enough to satisfy the committee's required drawdown limit. After costs, the combined out-of-sample max drawdown is -20.1% versus -22.0% for the SAA baseline, and TAA improves max drawdown in 4 of 4 test folds. The tactical rule that works best is a wider risk-off reweighting inside the +/-15% deviation band around the SAA weights. Even so, the Assignment 4 limits still leave too much structural risk in the portfolio, so the honest recommendation is to present this as the lowest-drawdown feasible overlay found rather than as a fully successful solution to the 18% or 13% target.

Strategic rebuild notes

The full paginated Zion pull contains 55,399 OHLCV rows. The six-asset overlap begins in 2010 when Bitcoin starts, while the staggered-history baseline becomes feasible in 2003 once REITs enter the universe and the IPS cap structure can be satisfied.

window	backtest_total_return_pa	backtest_volatility_pa	backtest_max_drawdown
63d	5.7%	8.2%	-26.3%
84d	5.9%	8.3%	-26.3%
126d	5.8%	8.2%	-26.3%
168d	5.2%	8.2%	-26.3%
252d	5.1%	8.2%	-26.3%